SkillNavora

Supplement B

Excel, SQL, business, governance and production topics

Excel automation, SQL extras, business problem-solving, requirements, governance and privacy, production Python, recommendation systems, APIs, Linux and documentation.

Part of the AI for Data Engineers (Data Science & AI) programme.

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Supplement B: Excel, SQL, Business, Governance and Production Python

Written for the same learner and in the same structure as Supplement A.

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B1 Excel Automation and Missing Excel Skills (course Module 1 gaps)

B1.1

Flash Fill, Text to Columns and standardising data

WHAT IS IT?

> Flash Fill detects a pattern from examples and fills the rest; Text to Columns splits one column into several using a delimiter or fixed width.

WHY IS IT IMPORTANT?

> Analysts spend much of their time cleaning data; these tools remove hours of manual work.

WHERE IS IT USED IN INDUSTRY?

Used in finance, HR and operations reporting to clean exports from systems.

HOW DOES IT WORK?

- Type the result for the first row, then press Ctrl+E (Flash Fill).
- Data > Text to Columns splits by comma, space or fixed width.
- Standardise case with PROPER, UPPER, LOWER and TRIM.

PRACTICAL EXAMPLE

> Full names 'Asha Rao' split into first and last name; phone numbers standardised to +91 format.

TOOL OR SOFTWARE

Microsoft Excel (Windows or Microsoft 365 web)

HANDS-ON EXERCISE

Clean a list of 200 customer names and phone numbers; document each step.

COMMON MISTAKES

Using Flash Fill on data with mixed patterns without checking the result.

INTERVIEW QUESTION

How would you split a column of 'City, State' values in Excel?

OFFICIAL AND FURTHER READING

- Microsoft: Flash Fill (official)

B1.2

Excel macros and VBA: recording and writing automation

WHAT IS IT?

> A macro is a recorded or written set of steps. VBA (Visual Basic for Applications) is the language behind macros in Excel.

WHY IS IT IMPORTANT?

> Repetitive monthly reports can run in seconds instead of hours, and a macro can be checked and reused.

WHERE IS IT USED IN INDUSTRY?

Common in finance, operations and back-office teams.

HOW DOES IT WORK?

- Developer tab > Record Macro, perform the steps, stop recording.
- Alt+F11 opens the VBA editor to read or write code.
- Save macro-enabled files as .xlsm and sign or review macros from unknown sources.

PRACTICAL EXAMPLE

> A macro formats a monthly sales sheet, removes blank rows and saves it as a PDF.

TOOL OR SOFTWARE

Microsoft Excel VBA editor

HANDS-ON EXERCISE

Record a macro that formats a table and then edit its code to add a loop over three sheets.

COMMON MISTAKES

Recording macros without reading the code; hard-coded cell addresses break when data moves.

INTERVIEW QUESTION

What is the risk of running macros from an unknown file?

OFFICIAL AND FURTHER READING

- Microsoft VBA documentation (official)

B1.3

Office Scripts and AI-assisted Excel formulas

WHAT IS IT?

> Office Scripts are TypeScript automations for Excel on the web. AI tools can draft formulas, but each formula must be checked.

WHY IS IT IMPORTANT?

> Office Scripts run in Microsoft 365 without desktop macros, and can be triggered by Power Automate.

WHERE IS IT USED IN INDUSTRY?

Used in Microsoft 365 teams to automate reports that refresh on a schedule.

HOW DOES IT WORK?

- Automate > New Script in Excel on the web.
- Scripts can be run by Power Automate on a schedule.
- AI-written formulas are tested on a sample before use.

PRACTICAL EXAMPLE

> A script creates a summary sheet each Monday; Power Automate runs it and emails the file.

TOOL OR SOFTWARE

Excel on the web, Power Automate

HANDS-ON EXERCISE

Write an Office Script that adds a totals row and run it.

COMMON MISTAKES

Trusting an AI formula without checking edge cases such as blanks and errors.

INTERVIEW QUESTION

What is an Office Script and how does it differ from VBA?

OFFICIAL AND FURTHER READING

- Office Scripts documentation (official)

B1.4

Excel business projects from the prompt (not yet in the course)

WHAT IS IT?

> Projects: customer complaint analysis, financial performance tracker, employee productivity dashboard, customer segmentation and sales performance dashboard.

WHY IS IT IMPORTANT?

> Portfolio projects show the skills to employers, with real business questions.

WHERE IS IT USED IN INDUSTRY?

Analyst roles in every industry.

HOW DOES IT WORK?

- Brief: business question, data, deliverable, metric.
- Clean, model, analyse, visualise, recommend.
- Write a one-page summary for a manager.

PRACTICAL EXAMPLE

> Complaint analysis: count complaints by category and month, flag repeat customers, recommend two fixes with expected impact.

TOOL OR SOFTWARE

Excel

HANDS-ON EXERCISE

Complete the complaint analysis brief with a sample dataset you create.

COMMON MISTAKES

Dashboards with no question or recommendation.

INTERVIEW QUESTION

How would you analyse customer complaints for a telecom company?

OFFICIAL AND FURTHER READING

- Microsoft Excel training (official)

B2 SQL Extras for Analysts and Data Engineers (course Module 2 gaps)

B2.1

Recursive CTEs and hierarchies

WHAT IS IT?

> A recursive CTE repeats a query until no new rows appear, used for hierarchies such as organisation charts or bill-of-materials.

WHY IS IT IMPORTANT?

> Hierarchical data appears in HR, product and supply chain systems.

WHERE IS IT USED IN INDUSTRY?

Org charts, category trees and supply chains.

HOW DOES IT WORK?

- WITH RECURSIVE tree AS (anchor UNION ALL recursive step).
- The recursive step joins the CTE to the table again.
- Always include a stop condition to avoid endless loops.

PRACTICAL EXAMPLE

> WITH RECURSIVE reports AS (SELECT id, manager_id, 1 AS level FROM staff WHERE manager_id IS NULL UNION ALL SELECT s.id, s.manager_id, r.level+1 FROM staff s JOIN reports r ON s.manager_id = r.id) SELECT * FROM reports;

TOOL OR SOFTWARE

PostgreSQL

HANDS-ON EXERCISE

Build the reporting-level query for a staff table of 20 people.

COMMON MISTAKES

Forgetting the stop condition.

INTERVIEW QUESTION

How would you find everyone who reports to a manager at any depth?

OFFICIAL AND FURTHER READING

- PostgreSQL WITH queries (official)

B2.2

Percentiles, medians and distribution queries

WHAT IS IT?

> Percentiles show the value below which a given share of rows fall. PERCENTILE_CONT interpolates; PERCENTILE_DISC picks an actual value.

WHY IS IT IMPORTANT?

> Medians and percentiles describe skewed business data (order values, delivery times) better than averages.

WHERE IS IT USED IN INDUSTRY?

Used for SLA reporting and pricing analysis.

HOW DOES IT WORK?

- PERCENTILE_CONT(0.5) WITHIN GROUP (ORDER BY amount) is the median in PostgreSQL.
- Check the database's exact syntax; platforms differ.

PRACTICAL EXAMPLE

> The 90th percentile of delivery time per city, to report SLA breaches.

TOOL OR SOFTWARE

PostgreSQL

HANDS-ON EXERCISE

Compute median and 90th percentile order value per region on a sample of 10,000 orders.

COMMON MISTAKES

Using AVG for skewed data without checking the median.

INTERVIEW QUESTION

Why report the median rather than the mean for delivery times?

OFFICIAL AND FURTHER READING

- PostgreSQL aggregate functions (official)

B2.3

Business SQL: cohort, funnel, churn, CLV and retention

WHAT IS IT?

> Cohort analysis groups customers by first purchase month and tracks their activity over time. Funnel analysis counts users at each step. Churn is the share lost in a period. CLV (customer lifetime value) estimates profit per customer over time.

WHY IS IT IMPORTANT?

> These are the core metrics in e-commerce, SaaS, banking and telecom analytics.

WHERE IS IT USED IN INDUSTRY?

Used by growth, marketing and customer success teams.

HOW DOES IT WORK?

- Cohort: first_month per customer, then count active customers by months since first purchase.
- Funnel: count distinct users per step, then divide.
- Churn: customers active last period and not this period, divided by active last period.
- CLV (simple): average order value x purchase frequency x expected years.

PRACTICAL EXAMPLE

> Cohort retention table: rows are first-purchase months, columns are months since first purchase, values are the share still buying.

TOOL OR SOFTWARE

PostgreSQL, sample e-commerce data

HANDS-ON EXERCISE

Build a cohort table for 12 months of sample orders and explain the drop after month 1.

COMMON MISTAKES

Counting orders instead of distinct customers.

INTERVIEW QUESTION

How do you calculate monthly churn with SQL?

OFFICIAL AND FURTHER READING

- Mode SQL tutorials: cohort analysis (official learning site)

B2.4

Incremental loading and data-quality SQL

WHAT IS IT?

> Incremental loading processes only new or changed rows using a watermark. Data-quality SQL tests check duplicates, nulls and orphans.

WHY IS IT IMPORTANT?

> Full reloads become too slow and expensive as data grows.

WHERE IS IT USED IN INDUSTRY?

Standard in data warehouse loading.

HOW DOES IT WORK?

- Store the last loaded timestamp.
- Select rows with updated_at greater than the watermark.
- Check: SELECT key, COUNT(*) FROM t GROUP BY key HAVING COUNT(*) > 1.

PRACTICAL EXAMPLE

> Load only orders updated since yesterday, then run a duplicate check on order_id.

TOOL OR SOFTWARE

PostgreSQL

HANDS-ON EXERCISE

Build an incremental load with a watermark table and two quality checks.

COMMON MISTAKES

Using a watermark based on load time instead of source update time.

INTERVIEW QUESTION

How do you load only new data into a warehouse table?

OFFICIAL AND FURTHER READING

- PostgreSQL SQL reference (official)

B2.5

Database platforms: SQL Server, BigQuery, Snowflake and Databricks SQL

WHAT IS IT?

> The same SQL idea works on several platforms; each has its own functions, pricing and performance features.

WHY IS IT IMPORTANT?

> Job descriptions name platforms; you should know the differences in syntax, pricing model and tuning.

WHERE IS IT USED IN INDUSTRY?

Enterprises use SQL Server; cloud analytics teams use BigQuery, Snowflake or Databricks SQL.

HOW DOES IT WORK?

- SQL Server: T-SQL, SSMS, indexes and columnstore.
- BigQuery: pay per bytes scanned; partition and cluster tables.
- Snowflake: virtual warehouses, pay per compute time.
- Databricks SQL: SQL over Delta tables.

PRACTICAL EXAMPLE

> The same daily revenue query runs on BigQuery with partitioned tables to cut scan cost.

TOOL OR SOFTWARE

BigQuery sandbox (check current terms), Snowflake trial (check terms)

HANDS-ON EXERCISE

Run one query on two platforms and compare how they are priced and tuned.

COMMON MISTAKES

Writing SELECT * on pay-per-scan platforms.

INTERVIEW QUESTION

How does BigQuery pricing change how you write queries?

OFFICIAL AND FURTHER READING

- BigQuery documentation (official)
- Snowflake documentation (official)

B3 Business Problem-Solving and Analytics Consulting (course Module 16 gap)

B3.1

Business KPIs, models and process mapping

WHAT IS IT?

> A KPI is a measured result tied to a goal. A business model explains how money is made; process mapping shows each step of a workflow.

WHY IS IT IMPORTANT?

> Analysts who know the business choose the right metric and find the real problem.

WHERE IS IT USED IN INDUSTRY?

Consultants and analysts in banking, retail and healthcare start here.

HOW DOES IT WORK?

- Start from the goal, then the KPI, then the driver metrics.
- Map the process step by step with owners and times.
- Mark the bottleneck and the cost of delay.

PRACTICAL EXAMPLE

> Loan approval takes 9 days; mapping shows 5 days in document collection. The KPI is time to approval; the driver is documents received on day 1.

TOOL OR SOFTWARE

Miro or draw.io for process maps

HANDS-ON EXERCISE

Map one process you know, with three KPIs and one bottleneck.

COMMON MISTAKES

Choosing vanity metrics.

INTERVIEW QUESTION

What is the difference between a KPI and a driver metric?

OFFICIAL AND FURTHER READING

- Atlassian: process mapping (official guide)

B3.2

Problem discovery: stakeholder interviews, Five Whys and fishbone analysis

WHAT IS IT?

> Stakeholder interviews collect pain points. Five Whys asks why repeatedly to reach a root cause. A fishbone diagram groups causes into categories (people, process, tools, data).

WHY IS IT IMPORTANT?

> Solving the symptom wastes money; the root cause is what to fix.

WHERE IS IT USED IN INDUSTRY?

Used in operations, quality and consulting projects.

HOW DOES IT WORK?

- Interview 5 to 8 people across roles.
- Write a problem statement: who, what, impact, how often.
- Five Whys: ask why until the cause is a process or system you can change.
- Fishbone: list causes under each category.

PRACTICAL EXAMPLE

> Problem: late deliveries. Why 1: stock is out. Why 2: forecast is wrong. Why 3: sales data is a week late. Root cause: weekly data load.

TOOL OR SOFTWARE

Whiteboard, Miro

HANDS-ON EXERCISE

Run a Five Whys on a problem from your work and write the root cause.

COMMON MISTAKES

Stopping at the first answer.

INTERVIEW QUESTION

How would you find the root cause of rising returns?

OFFICIAL AND FURTHER READING

- ASQ: root cause analysis (official)

B3.3

Hypotheses, analytical questions and recommendations

WHAT IS IT?

A hypothesis is a testable statement. Analysis answers a question with data; a recommendation says what to do and what result to expect.

WHY IS IT IMPORTANT?

Recommendations that are not tied to a question and a measure are ignored.

WHERE IS IT USED IN INDUSTRY?

Consultants and analysts present recommendations to executives.

HOW DOES IT WORK?

- Write the question, then the hypothesis, then the data needed.
- Test it with the data; state what would prove it wrong.
- Recommend an action with an owner and a measure.

PRACTICAL EXAMPLE

Hypothesis: a reminder email reduces missed appointments by 15%. Test with an A/B split over four weeks.

TOOL OR SOFTWARE

Excel, SQL, Python

HANDS-ON EXERCISE

Write three hypotheses for a shop's falling sales and the data test for each.

COMMON MISTAKES

Recommendations without expected impact.

INTERVIEW QUESTION

How do you turn an analysis into a recommendation?

OFFICIAL AND FURTHER READING

- Harvard Business Review: data-driven decisions (learning article)

B4 Requirements, Communication and Delivery

B4.1

Requirements, user stories, acceptance criteria and PRDs

WHAT IS IT?

> Requirements say what the solution must do. User stories describe value from the user's view. Acceptance criteria say how to test it. A PRD (product requirements document) gathers these.

WHY IS IT IMPORTANT?

> Clear requirements stop rework and disputes with stakeholders.

WHERE IS IT USED IN INDUSTRY?

Product, analytics and BI teams write these every week.

HOW DOES IT WORK?

- Story: As a [role], I want [need], so that [benefit].
- Acceptance: Given [context], when [action], then [result].
- Non-functional requirements: speed, security, accuracy, availability.

PRACTICAL EXAMPLE

> As an operations manager, I want a daily late-order list, so that I can call customers before 10:00. Given an order is late by 2 hours, then it appears on the list by 08:00.

TOOL OR SOFTWARE

Jira, Confluence

HANDS-ON EXERCISE

Write five user stories with acceptance criteria for a dashboard.

COMMON MISTAKES

Stories without testable criteria.

INTERVIEW QUESTION

What is an acceptance criterion?

OFFICIAL AND FURTHER READING

- Atlassian: user stories (official guide)

B4.2

Communicating insights and data storytelling

WHAT IS IT?

> Data storytelling states the main message first, shows the evidence in one chart, and ends with the decision needed.

WHY IS IT IMPORTANT?

> Executives decide from short, clear messages. Long reports go unread.

WHERE IS IT USED IN INDUSTRY?

Analysts, FDEs and consultants present to leaders weekly.

HOW DOES IT WORK?

- Start with the answer.
- One chart per message.
- Name the decision, the owner and the date.

PRACTICAL EXAMPLE

> Slide: Churn rose 4 points in March, driven by two plans. Decision: fix onboarding for those plans by May.

TOOL OR SOFTWARE

PowerPoint, Power BI

HANDS-ON EXERCISE

Turn a 10-slide report into a 3-slide summary with a decision.

COMMON MISTAKES

Showing every chart on the dashboard.

INTERVIEW QUESTION

How would you present bad news to a manager?

OFFICIAL AND FURTHER READING

- Storytelling with Data (official site)

B4.3

Agile and project delivery for analytics teams

WHAT IS IT?

> Agile delivers in short cycles with a backlog, sprints and reviews. Kanban shows flow on a board. Delivery management tracks scope, risk and dependencies.

WHY IS IT IMPORTANT?

> Analytics work is often unclear at the start; short cycles show progress and adjust scope.

WHERE IS IT USED IN INDUSTRY?

Used by data and product teams across industries.

HOW DOES IT WORK?

- Backlog ordered by value.
- Sprint of 1 to 2 weeks with a goal.
- Review with stakeholders; retrospect and improve.

PRACTICAL EXAMPLE

> A two-week sprint delivers the first version of a sales dashboard; the review finds two new metrics to add.

TOOL OR SOFTWARE

Jira, Trello

HANDS-ON EXERCISE

Plan a four-week analytics project in four sprints with goals.

COMMON MISTAKES

Planning without stakeholder reviews.

INTERVIEW QUESTION

How do you run an analytics project in sprints?

OFFICIAL AND FURTHER READING

- Scrum Guide (official)

B5 Data Governance, Privacy and Responsible AI (course Module 15 gap)

B5.1

Data ownership, stewardship and retention

WHAT IS IT?

> An owner decides who may use a dataset; a steward keeps it accurate. Retention sets how long data is kept.

WHY IS IT IMPORTANT?

> Clear ownership makes fixes happen; retention limits risk and cost.

WHERE IS IT USED IN INDUSTRY?

Required in banking, insurance and health data.

HOW DOES IT WORK?

- Each dataset has one owner and one steward.
- Retention period written in policy.
- Deletion runs on schedule.

PRACTICAL EXAMPLE

> Customer call recordings kept 90 days, then deleted by a scheduled job with a log.

TOOL OR SOFTWARE

Catalogue tools (see A4.3)

HANDS-ON EXERCISE

Write a data ownership sheet for five tables in a sales system.

COMMON MISTAKES

No owner for shared tables.

INTERVIEW QUESTION

Who should own customer data in a bank?

OFFICIAL AND FURTHER READING

- DAMA DMBOK overview (official body)

B5.2

Privacy: GDPR and India's Digital Personal Data Protection Act

WHAT IS IT?

> Privacy laws set rules for collecting, using, storing and deleting personal data. GDPR applies to EU residents' data; India's DPDP Act 2023 covers digital personal data in India.

WHY IS IT IMPORTANT?

> Breaches and fines follow poor privacy design. Analysts must minimise and protect personal data.

WHERE IS IT USED IN INDUSTRY?

Every customer-data project in India or Europe.

HOW DOES IT WORK?

- Collect only what is needed (data minimisation).
- Record consent and purpose.
- Give people access, correction and deletion rights where the law requires.
- Report breaches as the law requires.

PRACTICAL EXAMPLE

> A marketing dataset removes phone numbers not needed for the campaign and keeps a consent flag for each contact.

TOOL OR SOFTWARE

Data inventory sheet

HANDS-ON EXERCISE

Review a sample dataset and list the personal fields, purpose and retention for each.

COMMON MISTAKES

Copying production personal data into test files.

INTERVIEW QUESTION

What is data minimisation?

OFFICIAL AND FURTHER READING

- GDPR text (official EU site)
- MeitY: Digital Personal Data Protection Act (official)

B5.3

Data masking, row-level security and access control

WHAT IS IT?

> Masking hides sensitive values (for example showing the last four digits). Row-level security limits which rows a user sees.

WHY IS IT IMPORTANT?

> Analysts need data without exposing personal details.

WHERE IS IT USED IN INDUSTRY?

Used in banks, HR and healthcare analytics.

HOW DOES IT WORK?

- Mask in queries or in the database.
- Power BI and database RLS filter rows by user.
- Audit who accessed sensitive tables.

PRACTICAL EXAMPLE

> A regional manager sees only their region's rows in the Power BI report through RLS.

TOOL OR SOFTWARE

Power BI RLS, PostgreSQL row-level security

HANDS-ON EXERCISE

Set up a row filter for two regions and test it with two users.

COMMON MISTAKES

Relying on report-level hiding instead of security at the data layer.

INTERVIEW QUESTION

How do you stop an analyst from seeing salary data they should not see?

OFFICIAL AND FURTHER READING

- Power BI row-level security (official)

B5.4

Responsible AI, threat modelling and incident response

WHAT IS IT?

> Responsible AI covers fairness, transparency, human oversight and security. Threat modelling lists assets, threats and controls. Incident response plans how to act when something goes wrong.

WHY IS IT IMPORTANT?

> AI systems can leak data, be manipulated (prompt injection) or treat groups unfairly.

WHERE IS IT USED IN INDUSTRY?

Required for AI products in regulated sectors.

HOW DOES IT WORK?

- Check model performance by group.
- Threat model: data, prompts, tools, outputs.
- Plan detection, containment, fix and communication for incidents.

PRACTICAL EXAMPLE

> A chatbot leaks an internal price list through prompt injection; the incident plan disables the tool, rotates keys and updates guardrails.

TOOL OR SOFTWARE

OWASP Top 10 for LLM Applications

HANDS-ON EXERCISE

Write a threat model for an AI chatbot with five threats and controls.

COMMON MISTAKES

Treating guardrails as a complete solution.

INTERVIEW QUESTION

How would you test an AI assistant for prompt injection?

OFFICIAL AND FURTHER READING

- OWASP Top 10 for LLM Applications (official)
- NIST AI Risk Management Framework (official)

B6 Production Python for Automation (course Module 6 gaps)

B6.1

Automation: email, scheduled scripts and report generation

WHAT IS IT?

> Automation scripts run on a schedule, read files or APIs, produce reports and send them to people.

WHY IS IT IMPORTANT?

> Manual reports waste time and are error-prone.

WHERE IS IT USED IN INDUSTRY?

Common in every analytics team.

HOW DOES IT WORK?

- Use a scheduler (cron, Task Scheduler, Airflow).
- Send email with smtplib or a provider API; never store passwords in code.
- Log each run and alert on failure.

PRACTICAL EXAMPLE

> A script runs at 07:00, builds the daily sales PDF and emails it to managers.

TOOL OR SOFTWARE

Python, smtplib, cron

HANDS-ON EXERCISE

Automate a daily CSV summary and log each run.

COMMON MISTAKES

Hard-coded passwords.

INTERVIEW QUESTION

How would you schedule a Python report and handle failures?

OFFICIAL AND FURTHER READING

- Python smtplib (official)
- Python logging (official)

B6.2

Logging, type hints, packaging and code organisation

WHAT IS IT?

> Logging records what a program does. Type hints document expected types and let tools catch mistakes. Packaging makes code installable.

WHY IS IT IMPORTANT?

> Production code must be readable, observable and reusable.

WHERE IS IT USED IN INDUSTRY?

Expected in data engineering and software roles.

HOW DOES IT WORK?

- logging.basicConfig with levels (INFO, WARNING, ERROR).
- def total(values: list[float]) → float.
- pyproject.toml defines the package.

PRACTICAL EXAMPLE

> A pipeline module logs row counts and fails with a clear message on bad input.

TOOL OR SOFTWARE

Python logging, mypy (check docs)

HANDS-ON EXERCISE

Add logging and type hints to one of your scripts and run mypy.

COMMON MISTAKES

Logging secrets.

INTERVIEW QUESTION

What should a production log include?

OFFICIAL AND FURTHER READING

- Python logging HOWTO (official)

B7 Recommendation Systems (course Module 7 gap)

B7.1

Recommendation systems: collaborative and content-based

WHAT IS IT?

> Recommenders suggest items. Collaborative filtering uses patterns across users; content-based filtering uses item features; hybrid systems combine both.

WHY IS IT IMPORTANT?

> Recommendations drive sales and engagement in e-commerce, streaming and banking.

WHERE IS IT USED IN INDUSTRY?

Retail, media and fintech product teams.

HOW DOES IT WORK?

- Build a user-item matrix.
- Measure similarity between users or items.
- Evaluate with precision at k and hit rate on held-out interactions.

PRACTICAL EXAMPLE

> Users who bought the same two products get a suggestion of a third product bought by most of them.

TOOL OR SOFTWARE

scikit-learn, pandas, Surprise library (open source)

HANDS-ON EXERCISE

Build an item-based recommender on a sample ratings file and report precision at 5.

COMMON MISTAKES

Recommending items the user already owns.

INTERVIEW QUESTION

How would you recommend products to a new user with no history?

OFFICIAL AND FURTHER READING

- scikit-learn user guide (official)
- Surprise library docs (official)

B7.2

Regression metrics reference: MAE, MSE, RMSE and R-squared

WHAT IS IT?

> MAE is the average absolute error; MSE the average squared error; RMSE its square root, in the same units as the target; R-squared the share of variation explained.

WHY IS IT IMPORTANT?

> Choosing the right error metric matches the business cost of mistakes.

WHERE IS IT USED IN INDUSTRY?

Forecasting and pricing teams.

HOW DOES IT WORK?

- MAE is robust to outliers; RMSE punishes large errors.
- R-squared can mislead when models are very flexible.

PRACTICAL EXAMPLE

> A delivery-time model has RMSE 2.5 days: typical error is about two and a half days.

TOOL OR SOFTWARE

scikit-learn metrics

HANDS-ON EXERCISE

Compute MAE, RMSE and R-squared for a regression and explain each to a manager.

COMMON MISTAKES

Using R-squared alone to judge a model.

INTERVIEW QUESTION

When would you prefer MAE over RMSE?

OFFICIAL AND FURTHER READING

- scikit-learn regression metrics (official)

B8 APIs, OpenAPI, OAuth and Integration Depth (Module 13 gaps)

B8.1

OpenAPI and Swagger: describing APIs

WHAT IS IT?

> OpenAPI is a standard file (YAML or JSON) that describes an API's endpoints, parameters and responses. Swagger UI shows it as interactive documentation.

WHY IS IT IMPORTANT?

> Integrators and AI tools read OpenAPI files to call APIs correctly.

WHERE IS IT USED IN INDUSTRY?

Every API-first company publishes OpenAPI specs.

HOW DOES IT WORK?

- Paths and methods.
- Parameters and request bodies.
- Responses and schemas.
- Security schemes.

PRACTICAL EXAMPLE

> An OpenAPI file describes GET /orders?status=open and its response schema.

TOOL OR SOFTWARE

Swagger Editor

HANDS-ON EXERCISE

Write an OpenAPI description for two endpoints of your own app.

COMMON MISTAKES

Specs that are not kept in sync with the code.

INTERVIEW QUESTION

What is OpenAPI used for?

OFFICIAL AND FURTHER READING

- OpenAPI specification (official)

B8.2

OAuth 2.0 in depth and secure tokens

WHAT IS IT?

> OAuth 2.0 lets an app act for a user without the user's password. The app gets an access token with limited scopes; the user approves the access.

WHY IS IT IMPORTANT?

> Integrations with Google, Microsoft, Slack and CRMs use OAuth. Mistakes leak data.

WHERE IS IT USED IN INDUSTRY?

Every SaaS integration.

HOW DOES IT WORK?

- Authorisation code flow with PKCE for apps.
- Scopes limit what the token can do.
- Tokens expire; refresh tokens renew them.
- Never put tokens in URLs or logs.

PRACTICAL EXAMPLE

> A Slack bot asks for chat:write scope only; the token is stored in a secrets manager.

TOOL OR SOFTWARE

Postman OAuth helper

HANDS-ON EXERCISE

Get an OAuth token from a sandbox provider and call one endpoint with it.

COMMON MISTAKES

Using long-lived tokens in code.

INTERVIEW QUESTION

What is the difference between authentication and authorisation in OAuth?

OFFICIAL AND FURTHER READING

- OAuth 2.0 simplified (learning site)

B8.3

MCP security and business integration examples

WHAT IS IT?

> MCP servers expose tools to AI clients. Security needs scoped permissions, user consent for sensitive tools, and logs. Integration examples cover CRM, ERP, Jira, Slack and Notion.

WHY IS IT IMPORTANT?

> A badly scoped MCP tool can change business data.

WHERE IS IT USED IN INDUSTRY?

AI-enabled business tools in every sector.

HOW DOES IT WORK?

- Expose read tools first; add write tools with approval.
- Scope each tool to one system and user role.
- Log every tool call.

PRACTICAL EXAMPLE

> An MCP server lets the assistant read Jira tickets but needs approval to close them.

TOOL OR SOFTWARE

MCP documentation (official)

HANDS-ON EXERCISE

Design an MCP tool list for a support team with read and write tools marked.

COMMON MISTAKES

Giving an AI client admin tokens.

INTERVIEW QUESTION

How would you secure an MCP server that writes to a CRM?

OFFICIAL AND FURTHER READING

- Model Context Protocol (official)

B9 Linux and the Command Line (Part C gap)

B9.1

Linux and terminal essentials for data work

WHAT IS IT?

> Linux is the server operating system most data platforms run on. The command line lets you manage files, processes and services.

WHY IS IT IMPORTANT?

> Cloud servers, containers and CI runners are Linux; you need the terminal to debug them.

WHERE IS IT USED IN INDUSTRY?

Data engineering and DevOps roles.

HOW DOES IT WORK?

- Navigate: pwd, ls, cd, find, grep.
- Files: cat, head, tail -f, cp, mv, chmod.
- Processes: ps, top, kill.
- Pipes and redirection: |, >, >>.

PRACTICAL EXAMPLE

> tail -f pipeline.log | grep ERROR shows errors live as the pipeline runs.

TOOL OR SOFTWARE

Any Linux or WSL terminal

HANDS-ON EXERCISE

Find all ERROR lines in a log file from the last 1,000 lines and count them.

COMMON MISTAKES

Running commands from unknown sources.

INTERVIEW QUESTION

How would you find which process is using port 8080?

OFFICIAL AND FURTHER READING

- The Missing Semester (MIT, free)
- Linux command reference (free book)

B10 Technical Documentation and Production Troubleshooting (Part C gap)

B10.1

Technical documentation and runbooks

WHAT IS IT?

> Documentation explains what a system does, how to run it and how to fix it. A runbook is a step-by-step guide for common incidents.

WHY IS IT IMPORTANT?

> Handover and on-call work fail without documentation.

WHERE IS IT USED IN INDUSTRY?

Required in every production team.

HOW DOES IT WORK?

- README: purpose, setup, run, owners.
- Data dictionary: tables, columns, meanings.
- Runbook: symptom, check, fix, escalation.

PRACTICAL EXAMPLE

> A runbook for 'dashboard shows yesterday's numbers': check the pipeline run, re-run the failed task, notify the owner.

TOOL OR SOFTWARE

Markdown, Confluence

HANDS-ON EXERCISE

Write a runbook for one failure in your project.

COMMON MISTAKES

Documentation written once and never updated.

INTERVIEW QUESTION

What should a runbook contain?

OFFICIAL AND FURTHER READING

- Google SRE book: effective troubleshooting (free)

B10.2

Production troubleshooting for data pipelines and AI apps

WHAT IS IT?

> Troubleshooting finds why a live system fails: check what changed, read logs, reproduce, fix, verify and review.

WHY IS IT IMPORTANT?

> Stakeholders judge data teams on how fast and calmly they recover.

WHERE IS IT USED IN INDUSTRY?

On-call data engineers and FDEs.

HOW DOES IT WORK?

- Confirm impact and mitigate first (rerun, roll back, switch source).
- Check recent changes, logs and upstream freshness.
- Fix the root cause and add a test or alert.
- Write a short post-incident review.

PRACTICAL EXAMPLE

> A dashboard is wrong: an upstream schema change renamed a column; the fix maps the new name and a schema-change alert is added.

TOOL OR SOFTWARE

Airflow logs, dbt docs, cloud logs

HANDS-ON EXERCISE

Inject a broken schema change in a sample pipeline and write the incident note.

COMMON MISTAKES

Fixing without finding the cause.

INTERVIEW QUESTION

A pipeline failed at 3 am. What do you do first?

OFFICIAL AND FURTHER READING

- PagerDuty incident response guide (official)

B11 OLTP vs OLAP in Depth (Module 2 gap)

B11.1

OLTP and OLAP: two kinds of database work

WHAT IS IT?

> OLTP (online transaction processing) systems record many small, fast changes; OLAP (online analytical processing) systems answer big questions over lots of history.

WHY IS IT IMPORTANT?

> Running heavy reports on the app database slows the app for customers; designing an analytics model like an app database makes reports slow and complex.

WHERE IS IT USED IN INDUSTRY?

Every product company has both: the app's database (OLTP) and a warehouse or lakehouse for reporting (OLAP).

HOW DOES IT WORK?

- OLTP: many users, single-row reads and writes in milliseconds, normalised tables, row storage, strict transactions (ACID). Examples: PostgreSQL, MySQL, SQL Server for an app.
- OLAP: fewer users, queries over millions of rows, denormalised star schemas, columnar storage, data refreshed in batches or streams. Examples: BigQuery, Snowflake, Redshift, Fabric, Databricks SQL.
- Data moves from OLTP to OLAP through ETL or ELT pipelines, often with change data capture.
- Some databases handle both (HTAP), but most teams still separate the workloads.

PRACTICAL EXAMPLE

> A food-delivery app writes each order to PostgreSQL (OLTP). Every night the orders flow into BigQuery (OLAP), where a star schema answers 'revenue by city by week' in seconds.

TOOL OR SOFTWARE

PostgreSQL or MySQL for OLTP; BigQuery, Snowflake or DuckDB for OLAP practice

HANDS-ON EXERCISE

Take an orders schema in third normal form and redesign it as a star schema with one fact table and three dimensions. List which queries get simpler.

COMMON MISTAKES

Running month-end reports directly on the production database during business hours.

INTERVIEW QUESTION

What is the difference between OLTP and OLAP, and why do companies separate them?

OFFICIAL AND FURTHER READING

- Microsoft: OLTP and OLAP (official)

B12 Power BI in Depth (Module 3 gaps)

B12.1

Copilot in Power BI

WHAT IS IT?

> Copilot is Microsoft's AI assistant inside Power BI. It can suggest report pages, summarise visuals in plain language and help write DAX from a description.

WHY IS IT IMPORTANT?

> It speeds up first drafts, but its output must be checked like any junior analyst's work.

WHERE IS IT USED IN INDUSTRY?

Used by BI teams to draft report pages and narrative summaries for business users.

HOW DOES IT WORK?

- Copilot needs a paid Fabric or Premium capacity and an admin setting to be enabled; check current licensing in Microsoft's documentation.
- Prepare the model first: clear table and column names, descriptions and a proper date table make answers better.
- Ask for a report page or a summary of the current visuals in plain language.
- Review every visual, filter and number; edit or delete what is wrong.

PRACTICAL EXAMPLE

> A sales manager asks Copilot to 'create a page showing revenue by region and month with the top 5 products'. The analyst checks totals against a known measure before publishing.

TOOL OR SOFTWARE

Power BI Service or Desktop with Copilot enabled on a Fabric capacity

HANDS-ON EXERCISE

If you have access, ask Copilot for one report page and list every change you had to make. If not, write the same request as a specification and build it yourself.

COMMON MISTAKES

Publishing AI-generated visuals without checking filters, totals and date ranges.

INTERVIEW QUESTION

How would you use Copilot in Power BI responsibly?

OFFICIAL AND FURTHER READING

- Copilot in Power BI overview (official)

B12.2

Deployment pipelines: development, test and production

WHAT IS IT?

> Deployment pipelines move Power BI content through stages (development, test, production) so changes are checked before users see them.

WHY IS IT IMPORTANT?

> Editing the live report directly risks breaking dashboards that leaders use every day.

WHERE IS IT USED IN INDUSTRY?

Enterprise BI teams use stages with different data sources and reviewers.

HOW DOES IT WORK?

- Create a pipeline in the Power BI Service and assign a workspace to each stage (requires a Premium or Fabric capacity; check current requirements).
- Build in development, deploy to test and review with users, then deploy to production.
- Use deployment rules to point each stage at the right data source or parameter.
- Compare stages before deploying to see what changed.

PRACTICAL EXAMPLE

> A finance dashboard is changed in development, checked by two finance users in test against last month's closed numbers, then deployed to production.

TOOL OR SOFTWARE

Power BI Service deployment pipelines; Git integration in Fabric for version control

HANDS-ON EXERCISE

Write the deployment checklist you would use before promoting a report to production.

COMMON MISTAKES

Testing against different data than production uses, so issues appear only after release.

INTERVIEW QUESTION

How do you release changes to a widely used Power BI report safely?

OFFICIAL AND FURTHER READING

- Deployment pipelines (official)

B12.3

Incremental refresh

WHAT IS IT?

> Incremental refresh reloads only recent data (for example the last 10 days) and keeps older history, instead of reloading the whole table every time.

WHY IS IT IMPORTANT?

> Full refreshes of large tables are slow and can hit time limits; incremental refresh makes them fast and reliable.

WHERE IS IT USED IN INDUSTRY?

Used on large fact tables such as transactions, events and logs.

HOW DOES IT WORK?

- Create two parameters named RangeStart and RangeEnd (Date/Time type) in Power Query.
- Filter the date column using these parameters so the source can return only that range.
- Define the policy on the table: how much history to store and how much to refresh.
- Publish; the service partitions the table and refreshes only the recent partitions.
- Check that the filter folds to the source query, or refresh will still read everything.

PRACTICAL EXAMPLE

> A 5-year transactions table stores 5 years and refreshes the last 7 days each night, cutting refresh time from 40 minutes to 3.

TOOL OR SOFTWARE

Power BI Desktop (Power Query parameters) and Power BI Service

HANDS-ON EXERCISE

Set up RangeStart and RangeEnd on a sample table and write down the policy you would choose for a daily sales table.

COMMON MISTAKES

Filtering on a column that does not fold to the source, so the full table is still read.

INTERVIEW QUESTION

When and how would you use incremental refresh in Power BI?

OFFICIAL AND FURTHER READING

- Incremental refresh (official)

B12.4

Industry dashboards: HR, finance, marketing and operations

WHAT IS IT?

> Each business function tracks a standard set of measures. Knowing them lets you design a useful dashboard on the first attempt.

WHY IS IT IMPORTANT?

> Interview case studies and real projects usually ask for one of these four dashboards.

WHERE IS IT USED IN INDUSTRY?

Every company with a BI team builds these four dashboards.

HOW DOES IT WORK?

- HR: headcount, hires, attrition rate (leavers ÷ average headcount), time to hire, absenteeism, diversity mix, attrition by tenure band.
- Finance: revenue, gross margin %, operating expenses vs budget, EBITDA, cash flow, days sales outstanding, variance to forecast.
- Marketing: spend by channel, leads, cost per lead, conversion rate by funnel stage, customer acquisition cost, return on ad spend, campaign ROI.
- Operations: orders, on-time delivery %, average fulfilment time, backlog, defect or return rate, capacity utilisation, cost per order.
- Design for each: one headline row of 4–6 KPIs with targets, trends over time, one breakdown, and a drill-through to detail.

PRACTICAL EXAMPLE

> An operations dashboard shows on-time delivery at 91% against a 95% target, a weekly trend, a breakdown by city that reveals one late warehouse, and a drill-through to late orders.

TOOL OR SOFTWARE

Power BI or Tableau with a star schema per function

HANDS-ON EXERCISE

Pick one of the four functions and sketch the dashboard: KPIs with formulas, targets, charts and filters.

COMMON MISTAKES

Showing every available metric instead of the few that drive decisions.

INTERVIEW QUESTION

Which KPIs would you put on an HR attrition dashboard, and how would you calculate them?

OFFICIAL AND FURTHER READING

- Power BI design tips (official)

B13 AI for Data Analysis, Hands-on (Module 5 gaps)

B13.1

Anomaly detection with AI and Python

WHAT IS IT?

> Anomaly detection finds values that break the usual pattern, such as a sudden drop in orders or a spike in refunds.

WHY IS IT IMPORTANT?

> Catching problems early saves money; checking every chart by eye does not scale.

WHERE IS IT USED IN INDUSTRY?

Payments (fraud), e-commerce (sales drops), operations (sensor faults) and product analytics (broken tracking).

HOW DOES IT WORK?

- Start simple: compare each day with a rolling average and flag values more than 3 standard deviations away (rolling z-score).
- Account for seasonality: compare with the same weekday last week or use a model with weekly seasonality.
- For many variables, try an Isolation Forest (scikit-learn).
- In Power BI, line charts have a built-in Find anomalies option that explains possible drivers.
- Review flagged points with the business before alerting; tune thresholds to avoid alert fatigue.

PRACTICAL EXAMPLE

> Daily refunds average 120 with a standard deviation of 15; a day with 190 has z ≈ 4.7 and is flagged; the cause is a delayed delivery partner in one city.

TOOL OR SOFTWARE

pandas rolling().mean()/std(); scikit-learn IsolationForest; Power BI Find anomalies

HANDS-ON EXERCISE

On a daily metric, compute a 28-day rolling z-score in pandas, flag $|z| > 3$, and explain the top three flags.

COMMON MISTAKES

Alerting on every small spike, or ignoring weekly seasonality and flagging every Monday.

INTERVIEW QUESTION

How would you detect unusual drops in daily orders automatically?

OFFICIAL AND FURTHER READING

- scikit-learn outlier detection (official)
- Power BI anomaly detection (official)

B13.2

Forecasting with Excel, Power BI and Python

WHAT IS IT?

> Forecasting predicts future values, such as next quarter's sales, from past data.

WHY IS IT IMPORTANT?

> Plans for stock, staffing and budgets depend on forecasts; a simple, honest forecast with a range beats a precise-looking guess.

WHERE IS IT USED IN INDUSTRY?

Retail demand, call-centre staffing, revenue planning and inventory.

HOW DOES IT WORK?

- Plot the history and look for trend and seasonality.
- Excel: FORECAST.ETS and the Forecast Sheet give a seasonal forecast with a confidence range.
- Power BI: the Analytics pane adds a forecast to line charts.
- Python: statsmodels (Holt-Winters, SARIMA) or Prophet for trend, seasonality and holidays.
- Hold out the last few periods, compare forecast with actual, and report MAPE or MAE before using the forecast.

PRACTICAL EXAMPLE

> Monthly sales for 3 years are forecast for the next 6 months with Holt-Winters; on the 6-month holdout the MAPE is 7%, so the plan uses the forecast with a ±10% range.

TOOL OR SOFTWARE

Excel FORECAST.ETS; Power BI forecast; Python statsmodels or Prophet

HANDS-ON EXERCISE

Forecast 6 months of a public monthly series in Excel and in Python, and compare MAPE on a holdout.

COMMON MISTAKES

Reporting a single number without a range, or testing on the data used to fit the model.

INTERVIEW QUESTION

How would you forecast next quarter's demand, and how would you know if the forecast is good?

OFFICIAL AND FURTHER READING

- Excel FORECAST.ETS (official)
- statsmodels time series (official)

B13.3

AI-generated DAX, and how to check it

WHAT IS IT?

> Using ChatGPT, Claude, Gemini or Copilot to write DAX measures from a plain-language description.

WHY IS IT IMPORTANT?

> It saves time on syntax, but DAX depends on your model and filter context, so plausible-looking measures can be wrong.

WHERE IS IT USED IN INDUSTRY?

BI developers use AI to draft measures, then test them.

HOW DOES IT WORK?

- Give the AI your table and column names, relationships and the exact definition you need.
- Ask for the measure with VAR/RETURN and a short explanation.
- Test in a matrix against values you can calculate by hand (one month, one region).
- Check totals and blank handling; use DIVIDE for ratios.
- Keep the final measure documented in the model.

PRACTICAL EXAMPLE

> Prompt: 'Tables Sales[Amount], Sales[OrderDate] linked to 'Date'[Date]. Write a DAX measure for sales growth vs the same month last year, returning blank if last year is blank.' Then check one month by hand.

TOOL OR SOFTWARE

ChatGPT, Claude, Gemini or Copilot; Power BI Desktop for testing

HANDS-ON EXERCISE

Ask an AI for three measures (YTD, last-year growth, share of total), test each against hand calculations, and note any fixes.

COMMON MISTAKES

Pasting AI-generated DAX without testing it under different filters.

INTERVIEW QUESTION

How do you verify a DAX measure that an AI wrote for you?

OFFICIAL AND FURTHER READING

- DAX reference (official)

B13.4

Choosing AI tools for analysis: Julius AI, NotebookLM, Perplexity and general assistants

WHAT IS IT?

> A comparison of popular AI tools for analysis work, so you pick the right one for the job.

WHY IS IT IMPORTANT?

> Each tool is good at different things; using the wrong one wastes time or gives unreliable answers.

WHERE IS IT USED IN INDUSTRY?

Analysts, researchers and consultants use these for data questions, reading documents and web research.

HOW DOES IT WORK?

- Julius AI: upload a spreadsheet or CSV and ask questions; it writes and runs code to make tables and charts. Good for quick exploratory analysis.
- NotebookLM (Google): add your own documents and ask questions answered only from them, with citations. Good for reports, policies and research papers.
- Perplexity: answers questions by searching the web, with source links. Good for market and competitor research; check the sources.
- ChatGPT, Claude and Gemini: general assistants that also analyse uploaded files and write code. Good for mixed tasks.
- For every tool: do not upload confidential or personal data unless your company has approved it, and verify numbers against the source.

PRACTICAL EXAMPLE

> An analyst uses Julius AI to explore a sales CSV, NotebookLM to summarise three internal research reports, and Perplexity to check competitor pricing news, citing every source.

TOOL OR SOFTWARE

Julius AI, NotebookLM, Perplexity, ChatGPT, Claude, Gemini (features and plans change; check each site)

HANDS-ON EXERCISE

Run the same question on one dataset or document in two tools and compare accuracy, citations and time taken.

COMMON MISTAKES

Trusting answers without checking sources, or uploading private data to an unapproved tool.

INTERVIEW QUESTION

Which AI tool would you use to analyse a CSV, to read a set of PDFs, and to research a market, and why?

OFFICIAL AND FURTHER READING

- NotebookLM (official)
- Perplexity (official)
- Julius AI (official)

B14 Business Statistics Cases (Module 6 gap)

B14.1

Worked case: revenue forecasting for a business plan

WHAT IS IT?

> Using simple statistics to turn past revenue into a forecast with a range that a business can plan with.

WHY IS IT IMPORTANT?

> Finance and leadership decisions need a forecast with honest uncertainty, not a single hopeful number.

WHERE IS IT USED IN INDUSTRY?

Annual plans, investor updates and budget reviews.

HOW DOES IT WORK?

- Collect 24–36 months of revenue; remove one-off events or mark them.
- Split revenue into drivers: customers × orders per customer × average order value.
- Estimate each driver's trend and seasonality (month-of-year index).
- Combine into a base case, then build a low and a high case from the historical variation (for example ±1 standard deviation of monthly growth).
- Check the method by forecasting last year from the year before and measuring MAPE.

PRACTICAL EXAMPLE

> Average monthly growth was 2.0% with a standard deviation of 1.5 points; the plan uses 2.0% (base), 0.5% (low) and 3.5% (high), so monthly revenue of ₹1.2 crore today becomes about ₹1.27 crore (low), ₹1.52 crore (base) or ₹1.81 crore (high) in 12 months (1.2×1.005^{12}, 1.2×1.02^{12}, 1.2×1.035^{12}).

TOOL OR SOFTWARE

Excel or Google Sheets; Python pandas for the backtest

HANDS-ON EXERCISE

Build a driver-based forecast for a public company's quarterly revenue with base, low and high cases, and backtest one year.

COMMON MISTAKES

Forecasting total revenue directly without understanding which driver moves it.

INTERVIEW QUESTION

How would you forecast next year's revenue for a subscription business?

OFFICIAL AND FURTHER READING

- Forecasting: Principles and Practice (free textbook)

B14.2

Worked case: risk analysis with expected value and simulation

WHAT IS IT?

> Quantifying risk by combining how likely something is with how much it would cost, and simulating many possible outcomes.

WHY IS IT IMPORTANT?

> Turns vague worries into numbers that can be compared and prioritised.

WHERE IS IT USED IN INDUSTRY?

Credit risk in lending, supply risk in operations, project risk and insurance.

HOW DOES IT WORK?

- List risks; estimate probability and impact for each.
- Expected loss = probability × impact; rank risks by expected loss.
- Where inputs are uncertain, use a range and run a Monte Carlo simulation (thousands of random draws).
- Report the average and a bad-case percentile (for example the 95th) of the simulated loss.
- Revisit estimates as real data arrives.

PRACTICAL EXAMPLE

> A supplier delay has a 10% chance and ₹20 lakh impact (expected loss ₹2 lakh); a data breach has a 1% chance and ₹5 crore impact (expected loss ₹5 lakh), so the breach is the higher priority.

TOOL OR SOFTWARE

Excel RAND() with Data Tables; Python NumPy for simulation

HANDS-ON EXERCISE

Simulate 10,000 outcomes for a project with three uncertain cost items and report the mean and 95th percentile total cost.

COMMON MISTAKES

Ranking risks by impact only and ignoring probability, or presenting simulated numbers as certain.

INTERVIEW QUESTION

How would you compare and prioritise business risks with numbers?

OFFICIAL AND FURTHER READING

- NumPy random sampling (official)